

Lecture-9

The Application Layer

Dr. Md. Imdadul Islam

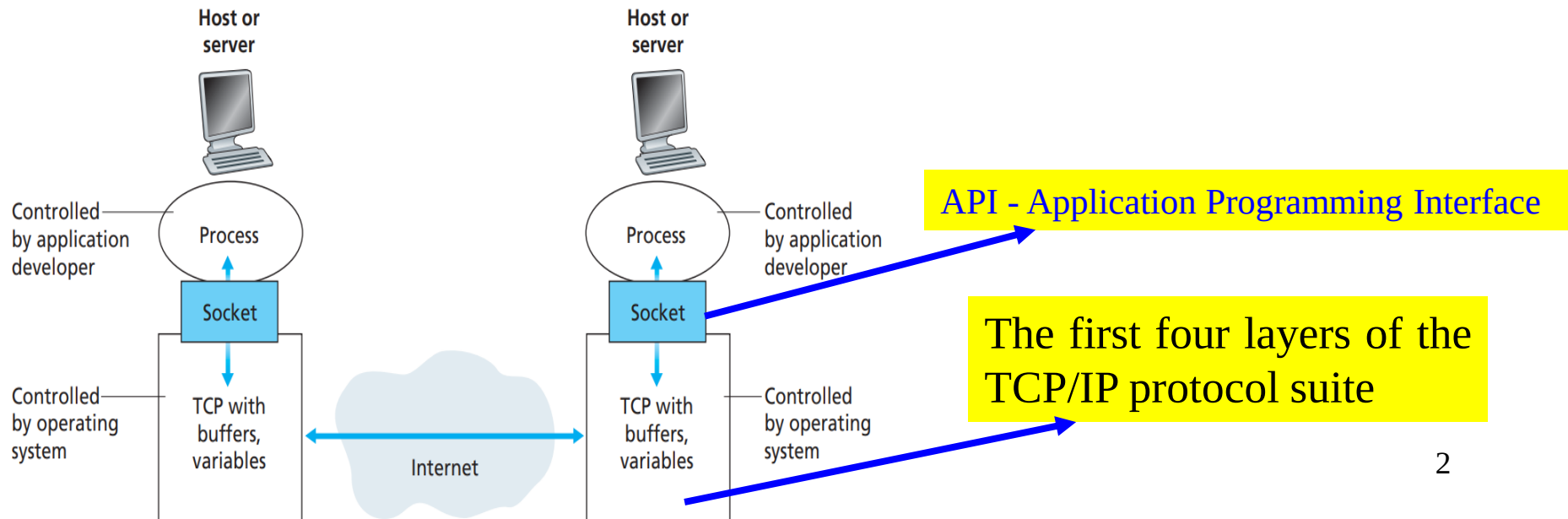
Professor, Department of Computer Science and
Engineering

Jahangirnagar University

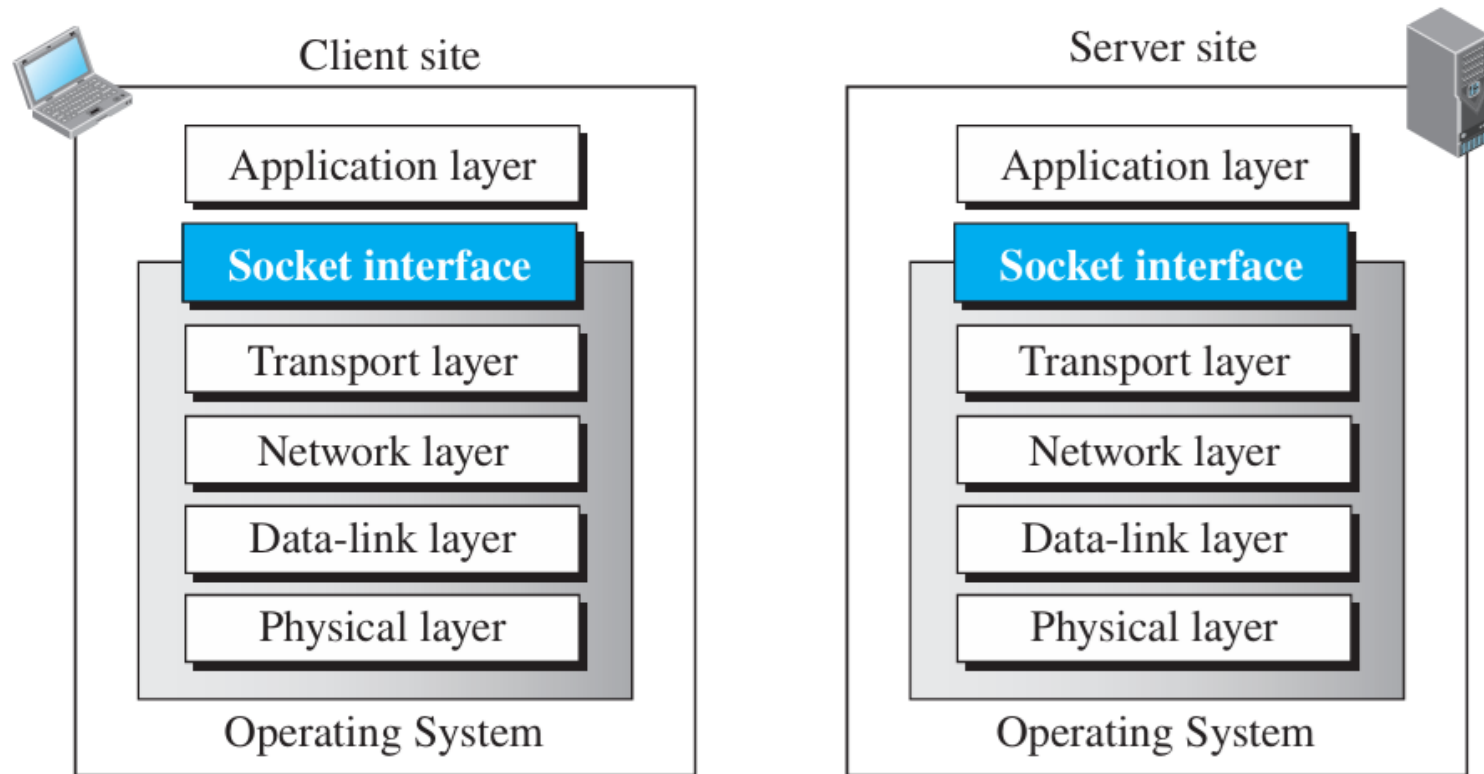
<https://www.juniv.edu/teachers/imdad>

Application Programming Interface

- ✓ If we need a **process** to be able to communicate with another **process**, we need a new **set of instructions** between two entities.
- ✓ In this case, one of the entities is the process at the application layer and the other is the operating system that encapsulates the first four layers of the TCP/IP protocol suite.
- ✓ A computer manufacturer needs to build the first four layers of the suite/stack in the operating system and include an API/ socket interface.

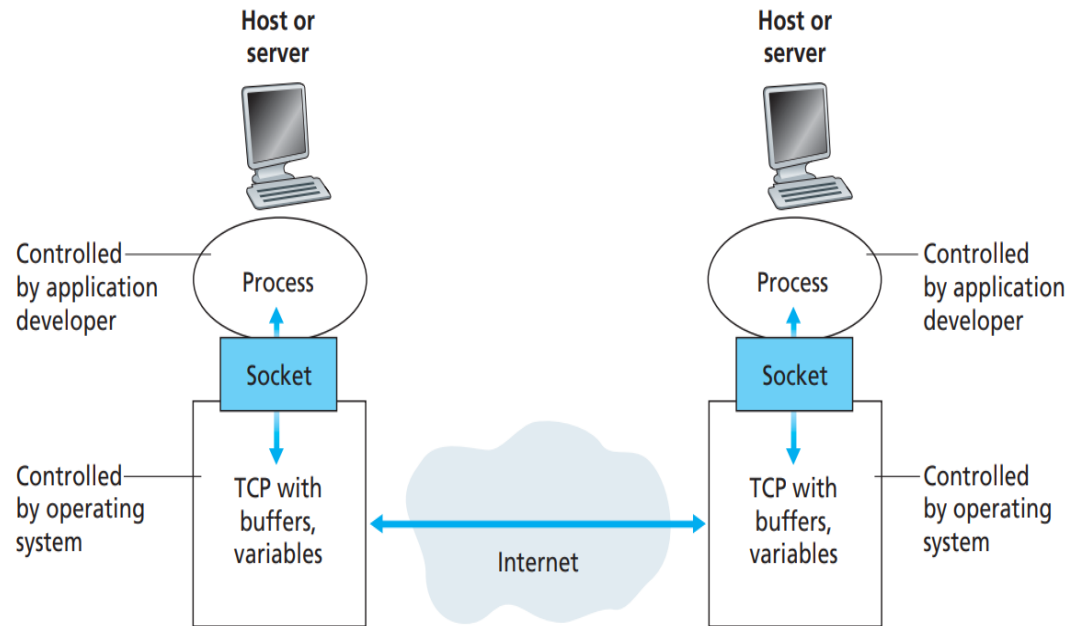


- ✓ The process at application layer tells the lowest four layers of the TCP/IP protocol suite to open the connection, send and receive data from the other end, and close the connection.
- ✓ The sending and receiving messages through the Internet is done by the first four layer. Most important API is socket interface.



Position of the socket interface

- ✓ Consider an analogy to help us understand processes and sockets. A process is analogous to a house and its socket is analogous to its door.



- ✓ When a process wants to send a message to another process on another host, it shoves (or push) the message out its door (socket). This sending process assumes that there is a transportation infrastructure from sending door to the door of the destination process.
- ✓ Once the message arrives at the destination host, the message passes through the receiving process's door (socket), and the receiving process then acts on the message.

Addressing Processes

- ✓ A process running on sending host needs the full address of the process running on the receiving host. To identify each process, two pieces of information is needed : (1) the address of the host (IP address) and (2) an identifier called port number that specifies the application process.
- ✓ This port number is needed because in general a host runs many network applications.
- ✓ For example, a Web server is identified by port number 80. A mail server process (using the SMTP protocol) is identified by port number 25.

Client and Server Processes

- ✓ A network application consists of pairs of processes that send messages to each other over a network. For example, in the Web application a **client browser process** exchanges messages with a **Web server process**.
- ✓ In a P2P (peer-to-peer) file-sharing system, a file is transferred from a process in one peer to a process in another peer. In P2P one of the two processes as the client and the other process as the server.

Application layer

- ✓ Users are familiar with several network applications: e-mail, World Wide Web, DNS, remote file access (Telnet service) and file transfer etc.
- ✓ Each application can be considered as a **set of programs** (or program) that run on different **end systems** and communicate with each other over the network.
- ✓ When a **new application is developed**, it is written for end machine but it does not need to be rewritten for networking devices, such as routers, that function at the network layer.

✓ This layer also contains **support protocols**, to allow the applications to function.

✓ An **application layer protocol** is associated with each **application**. Consider World Wide Web browsers that are available (Firefox, Safari, Internet Explorer etc.). You can use any one of these application programs to access a particular site on the Web is because they all use the same application layer protocol: HTTP (Hyper Text Transport Protocol).

Q. Give the idea of network application and application protocol.

This lecture will deal with open protocols:

- ✓ **Domain Name System (DNS)**
- ✓ **SMTP-The Simple Mail Transfer Protocol**
- ✓ **The World Wide Web**
- ✓ **URLs-Uniform Resource Locators**
- ✓ **Web Caching (Proxy Server)**
- ✓ **Telnet and Secure Shell**
- ✓ **File Transfer Protocol (FTP)**
- ✓ **Video streaming systems**
- ✓ **Network Management**

**Proprietary protocols:
Skype, Zoom**

Domain Name System (DNS)

- ✓ In Internet a host, web pages, mailboxes, web server and other resources are identified by **network addresses** (IP address) of the computers on which they are stored, these addresses are hard for people to remember.
- ✓ Initially a JU's Web pages were at 128.111.24.41 and if the university moves the Web server to a **different machine with a different IP address**, everyone needs to be told the new IP address.
- ✓ Consequently, high-level, readable names were introduced like: *www.juniv.edu/department/cse* against the JU website regardless of its IP address.
- ✓ Nevertheless, the network itself understands only numerical addresses, so some mechanism is required to convert the **ASCII strings to network addresses**.

✓ DNS is a tree type global directory (**like distributed database**) that translates **machine** or **domain names** to **numerical IP address**.

✓ DNS is an application layer protocol, and every **Internet service provider (ISP)** has a **DNS server**. In the normal mode of operation, a hosts send UDP (connection less) queries to a DNS server. The DNS server either replies or directs the queries to another server.

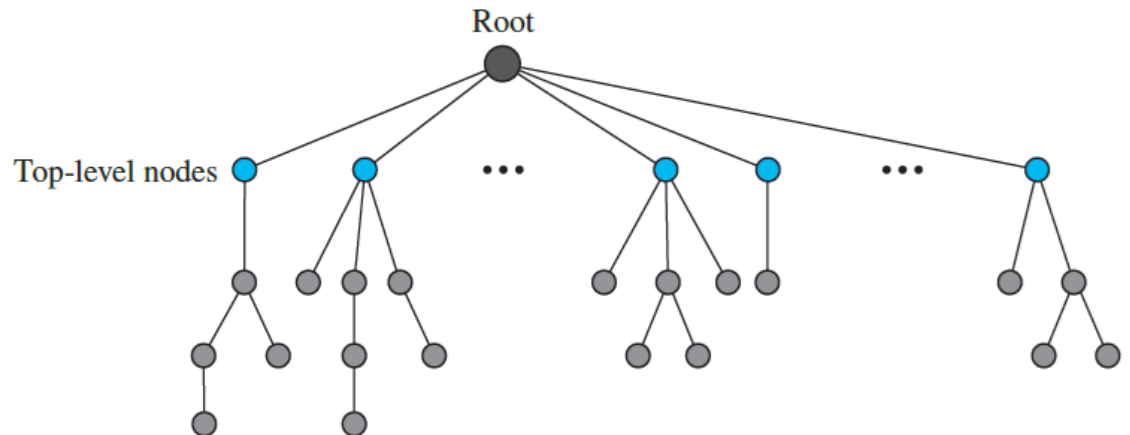
✓ A **distributed database** is a database in which storage devices are not all attached to a common processing unit such as the CPU, and which is controlled by a distributed database management system (together sometimes called a distributed database system). It may be stored in multiple computers, located in the same physical location; or may be dispersed over a network of interconnected computers.

Q. Why DNS server is used?

The DNS Name Space

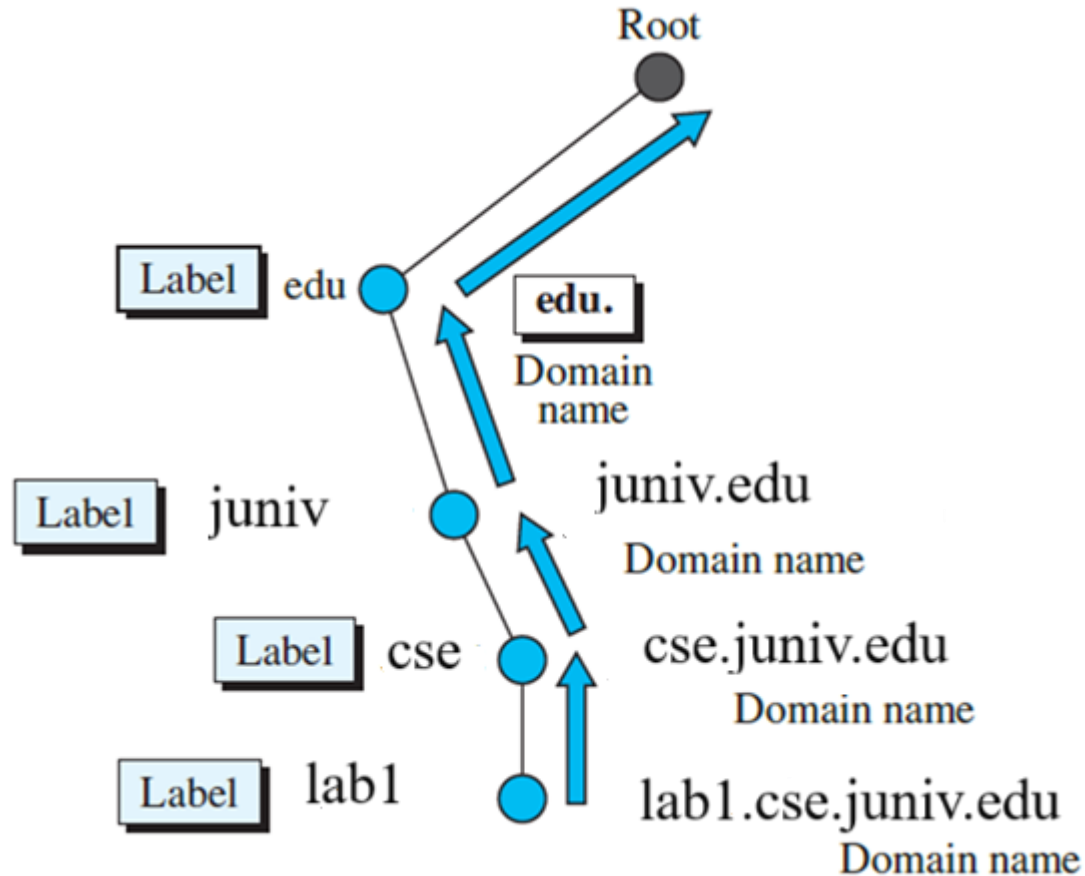
- ✓ In theory at least, a single name server could contain the entire DNS database and respond to all queries about it. In practice, this server would be so overloaded as to be useless.
- ✓ To avoid above problem, the DNS name space is divided into **nonoverlapping zones**.
- ✓ Conceptually, the Internet is divided into over 250 top-level domains (first level of tree), where each domain covers many hosts. Each domain is partitioned into sub-domains, and these are further partitioned, and so on.

Portion of the hierarchy of DNS servers.



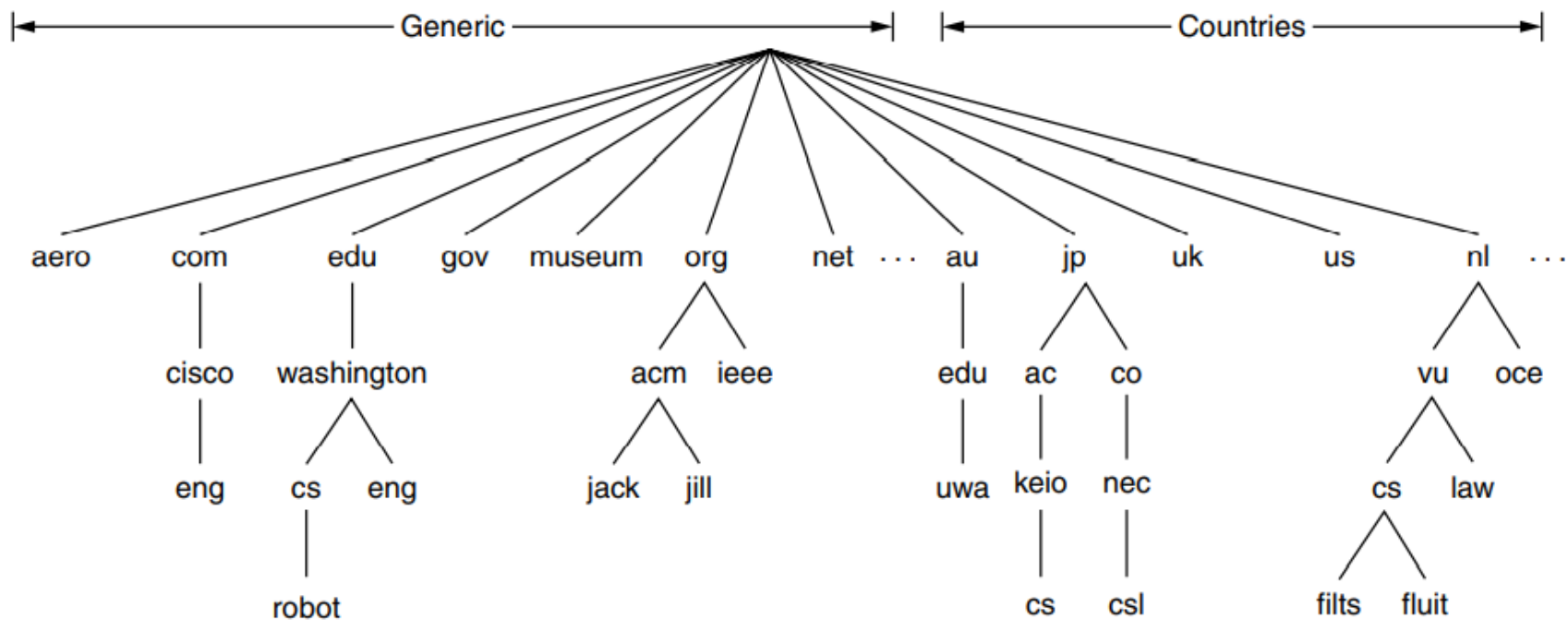
Domain Name

Each node in the tree has a domain name. A full **domain name** is a sequence of labels separated by dots (.). The domain names are always read from the node up to the root.



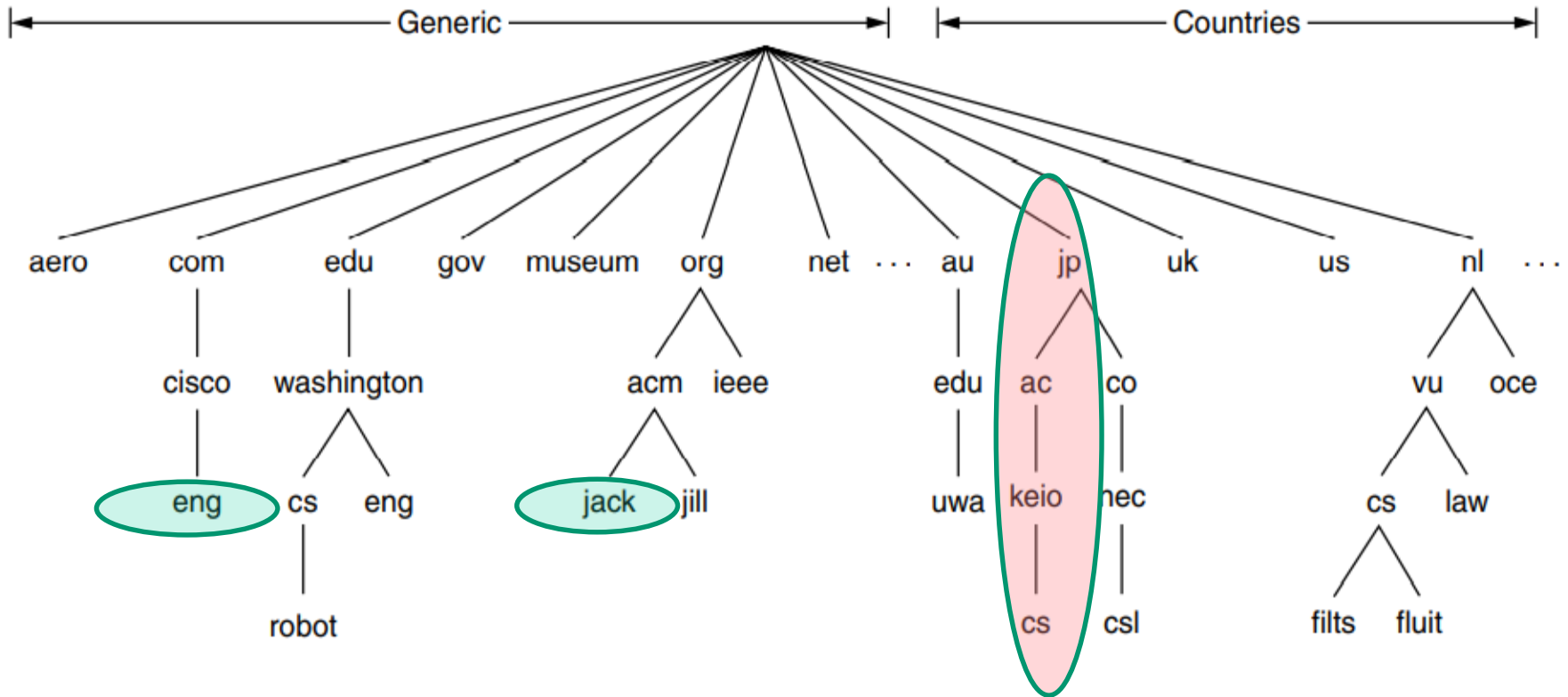
Domain names and labels

- ✓ The top-level domains come in two flavors: **generic** and **countries**.
- ✓ The original generic domains were *com* (commercial), *edu* (educational institutions), *gov* (the U.S. Federal Government), *int* (certain international organizations), *mil* (the U.S. armed forces), *net* (network providers), and *org* (nonprofit organizations).
- ✓ The country domains include one entry for every country.



✓ For the Internet, the top of the naming hierarchy is managed by an organization called **ICANN** (Internet Corporation for Assigned Names and Number).

- ✓ A leaf domain may contain a single host, or it may represent a company and contain thousands of hosts.



The domain names are written with the local label first and the top domain last.

cs.keio.ac.jp (CS department of Keio University, in Japan)

Fig.1 A portion of the Internet domain name space¹⁵

- ✓ Each domain controls the sub-domains under it.
- ✓ For example, Japan (*jp*) has domains *ac.jp* and *co.jp* that mirror *edu* and *com*. The Netherlands does not make this distinction and puts all organizations directly under *nl*.
- ✓ Thus, all three of the following are university computer science departments:
 1. *cs.washington.edu* (University of Washington, in the U.S.).
 2. *cs.vu.nl* (Vrije University, in The Netherlands).
 3. *cs.keio.ac.jp* (Keio University, in Japan).

Example-1

The domain name *cs.purdue.edu* contains three labels: *cs*, *purdue* and *edu*. It gives the domain name for Computer Science Department at Purdue University. Again *purdue.edu* gives the domain name of Purdue University.

Example-2

Let the host *private* (whose machine/domain name is *private.pbhs.brevard.k12.fl.us* and IP address is *204.128.68.25*) wants to connect the Web server *raider.rhs.brevard.k12.fl.us*. The web browser on *private* places a DNS query to the DNS server for *brevard.k12.fl.us*, which is authorized for the *brevard.k12.fl.us* domain. This DNS server looks up the information in its database and returns the IP address.

raider.rhs.brevard.k12.fl.us

DNS server for
brevard.k12.fl.us

pbhs.brevard.k12.fl.us

rhs.brevard.k12.fl.us

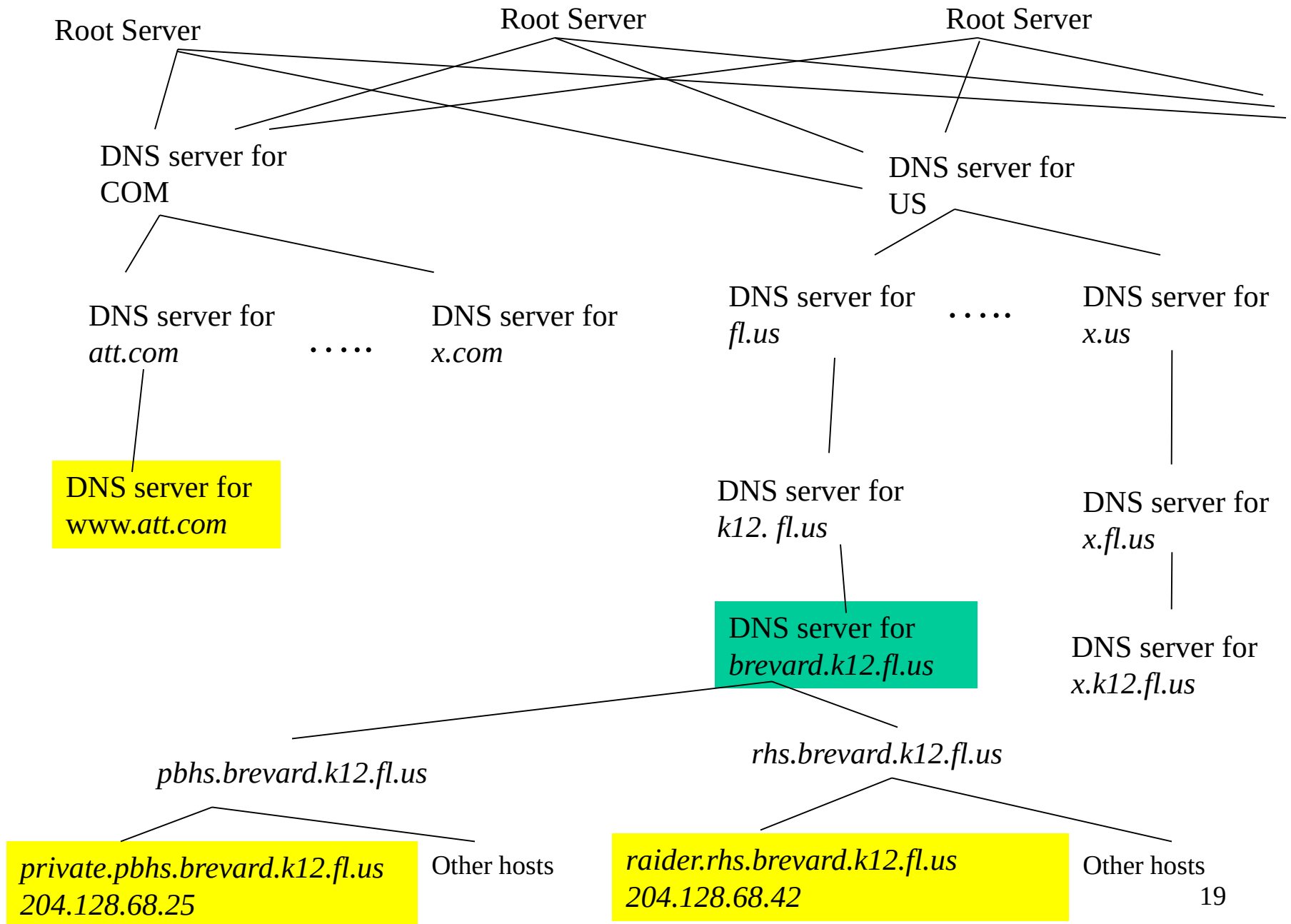
private.pbhs.brevard.k12.fl.us
204.128.68.25

Other hosts

raider.rhs.brevard.k12.fl.us
204.128.68.42

Other hosts

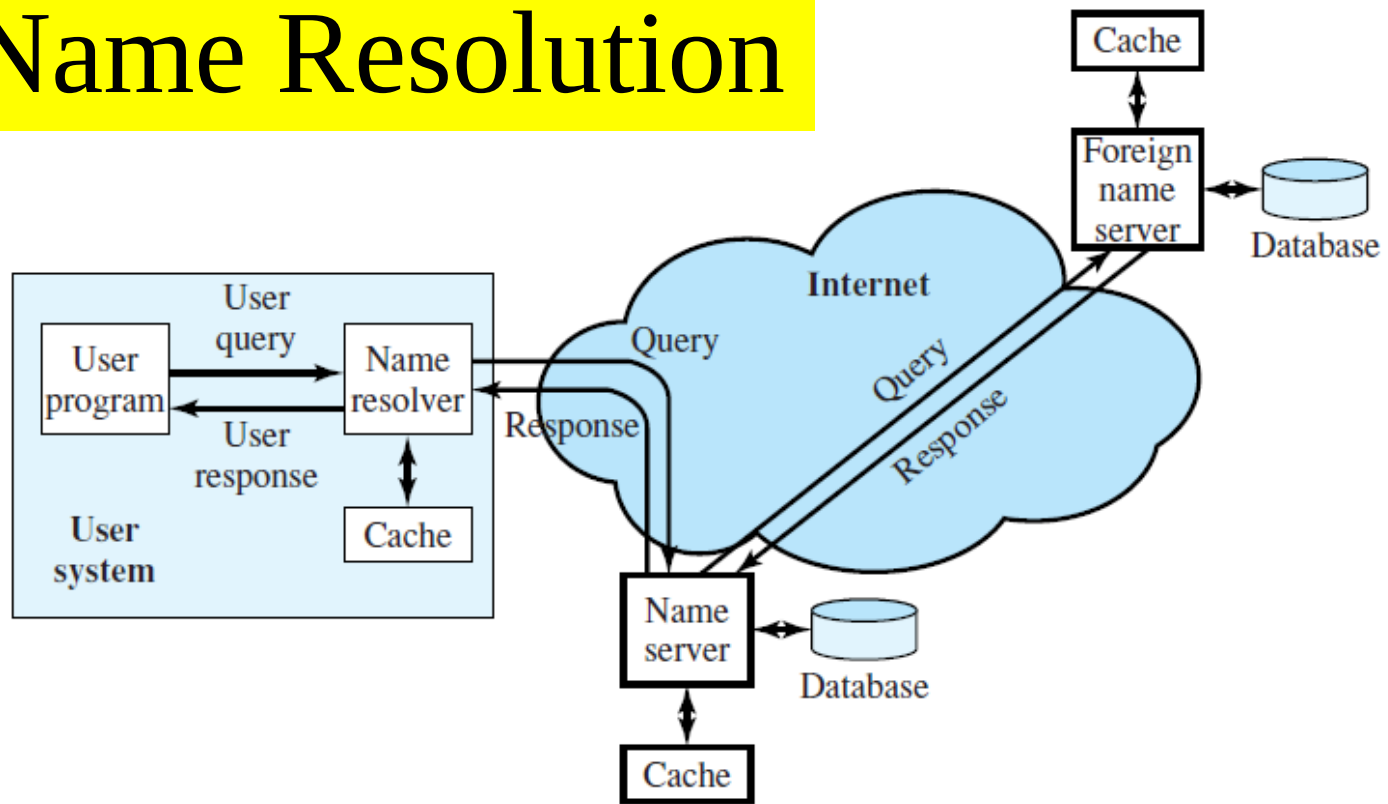
A **Web server** is a program that uses HTTP (Hypertext Transfer Protocol) to serve the files that form **Web** pages to users, in response to their requests, which are forwarded by their computers' HTTP clients. Dedicated computers and appliances may be referred to as **Web servers** as well.



✓ Now suppose that user wants to connect the web server *www.att.com*. A similar query is made to the *brevard.k12.fl.us* DNS server. This time it does not have any information about *www.att.com* domain and hence cannot resolve the name. Each DNS server has the address of other name server including at least one root server.

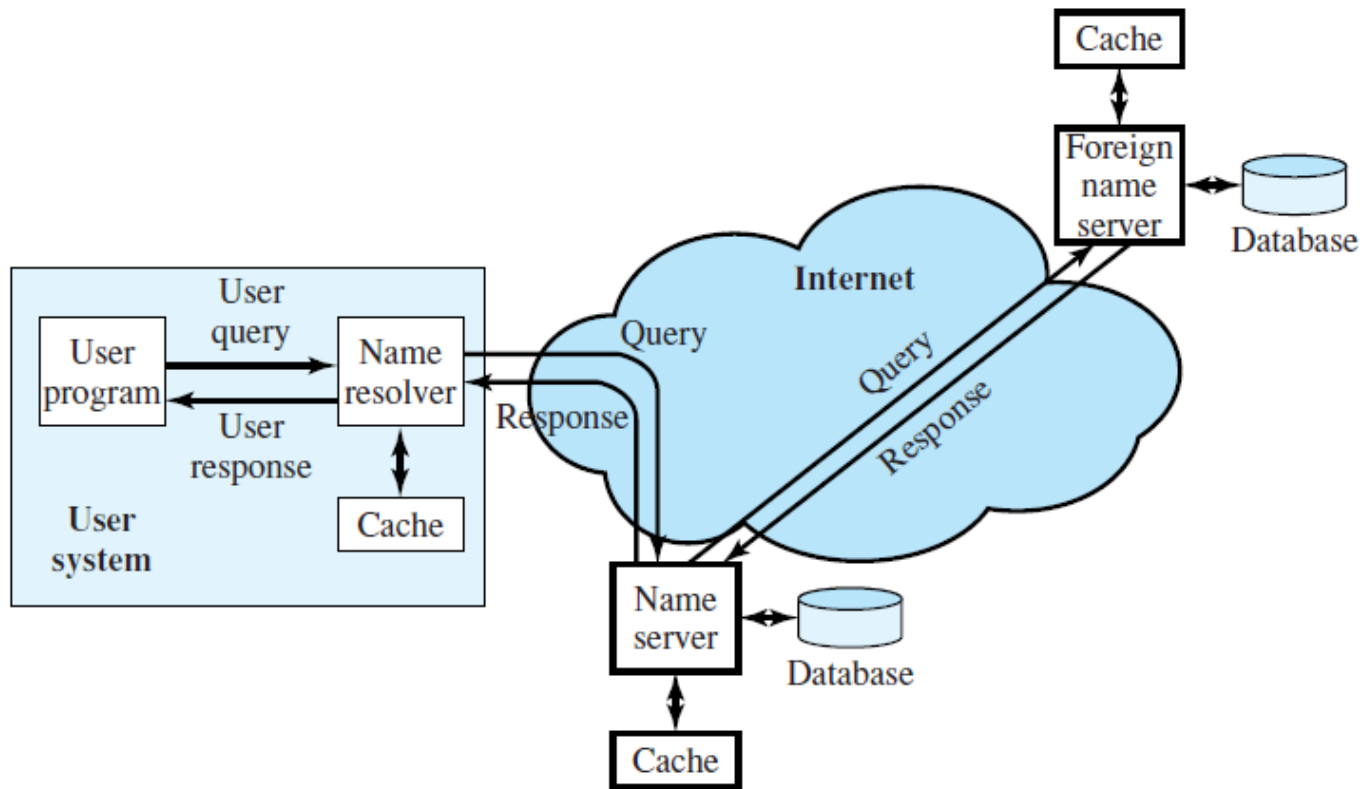
✓ If DNS server can not resolve the name it replies by specifying the name server that should be connected next. Eventually a root server gets involved. Root server maintain information about all the authoritative name server for each top level domain. A root server will eventually provide it with the *att.com* DNS server's address. The *brevard* name server then contacts the *att.com* name server, which will return the address 192.20.3.54

DNS Name Resolution



DNS operation typically includes the following steps (Figure above):

1. A user program requests an IP address for a domain name.
2. The local name server checks to see if the name is in its local database or cache, and, if so, returns the IP address to the requestor.
3. Otherwise, the name server queries other available name servers, starting down from the DNS tree or as high up the tree as possible.



4. When a response is received at the local name server, it stores the name/address mapping in its local cache and may maintain this entry for the amount of time specified in the **time to live** field of the retrieved RR (*resource records*).

5. The user program is given the IP address or an error message.

Q. Give the steps DNS name resolution.

Resource Records

Each resource record of DNS database has is a five-tuple:

Domain_name **Time_to_live** **Class** **Type** **Value**

- ✓ The *Domain_name* tells the domain to which this record applies.
- ✓ The *Time_to_live* field gives an indication of how stable the record is.
- ✓ The third field is the *Class*. For Internet information, it is always *IN*. For non-Internet information, other codes can be used, but in practice, these are rarely seen.

- ✓ The *Type* field tells what kind of record this is. The most important types are listed in Table. below.
- ✓ Finally, we have the *Value* field. This field can be a number, a domain name, or an ASCII string.

<i>Type</i>	<i>Interpretation of value</i>
A	A 32-bit IPv4 address (see Chapter 7)
NS	Identifies the authoritative servers for a zone
CNAME	Defines an alias for the official name of a host
SOA	Marks the beginning of a zone
MX	Redirects mail to a mail server
AAAA	An IPv6 address (see Chapter 7)

The principal DNS resource record types

Example-1 What do you mean by record:

www.ait.ac.th 86400 IN A 192.168.30.44

Ans.
DNS server will provide IP address 192.168.30.44 against *www.ait.ac.th*

<i>Type</i>	<i>Interpretation of value</i>
A	A 32-bit IPv4 address (see Chapter 7)
NS	Identifies the authoritative servers for a zone
CNAME	Defines an alias for the official name of a host
SOA	Marks the beginning of a zone
MX	Redirects mail to a mail server
AAAA	An IPv6 address (see Chapter 7)

Domain_name Time_to_live Class Type Value

Example-2 What do you mean by record:

ait.ac.th 86400 IN MX kuddus.ait.ac.th

If someone wants to send e-mail to, for example, *bill@ait.ac.th*, the sending host needs to find a mail server at *ait.ac.th* that is willing to accept e-mail. The *MX* record can provide this information.

ait.ac.th 86400 IN MX kuddus.ait.ac.th

The mail *bill@ait.ac.th* will be accepted by the **Mail server** with DNS of *kuddus.ait.ac.th*

Type	Interpretation of value
A	A 32-bit IPv4 address (see Chapter 7)
NS	Identifies the authoritative servers for a zone
CNAME	Defines an alias for the official name of a host
SOA	Marks the beginning of a zone
MX	Redirects mail to a mail server
AAAA	An IPv6 address (see Chapter 7)

Example-3

CNAME records allow aliases to be created.

cs.mit.edu 86400 IN CNAME lcs.mit.edu

Type	Meaning	Value
SOA	Start of authority	Parameters for this zone
A	IPv4 address of a host	32-Bit integer
AAAA	IPv6 address of a host	128-Bit integer
MX	Mail exchange	Priority, domain willing to accept email
NS	Name server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
SPF	Sender policy framework	Text encoding of mail sending policy
SRV	Service	Host that provides it
TXT	Text	Descriptive ASCII text

Example-4 What do you mean by the records:

juniv.edu 86400 IN TXT “Jahangirnagar University”
juniv.edu 86400 IN A 192.168.2.45

; Authoritative data for cs.vu.nl

cs.vu.nl. 86400 IN SOA star boss (9527,7200,7200,241920,86400)
cs.vu.nl. 86400 IN TXT "Divisie Wiskunde en Informatica."
cs.vu.nl. 86400 IN TXT "Vrije Universiteit Amsterdam."
cs.vu.nl. 86400 IN MX 1 zephyr.cs.vu.nl.
cs.vu.nl. 86400 IN MX 2 top.cs.vu.nl.

flits.cs.vu.nl. 86400 IN HINFO Sun Unix
flits.cs.vu.nl. 86400 IN A 130.37.16.112
flits.cs.vu.nl. 86400 IN A 192.31.231.165
flits.cs.vu.nl. 86400 IN MX 1 flits.cs.vu.nl.
flits.cs.vu.nl. 86400 IN MX 2 zephyr.cs.vu.nl.
flits.cs.vu.nl. 86400 IN MX 3 top.cs.vu.nl.
www.cs.vu.nl. 86400 IN CNAME star.cs.vu.nl
ftp.cs.vu.nl. 86400 IN CNAME zephyr.cs.vu.nl

rowboat IN A 130.37.56.201
IN MX 1 rowboat
IN MX 2 zephyr
IN HINFO Sun Unix

little-sister IN A 130.37.62.23
IN HINFO Mac MacOS

laserjet IN A 192.31.231.216
IN HINFO "HP Laserjet IIISi" Proprietary

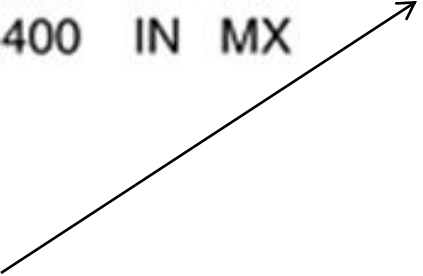
Type	Meaning	Value
SOA	Start of Authority	Parameters for this zone
A	IP address of a host	32-Bit integer
MX	Mail exchange	Priority, domain willing to accept e-mail
NS	Name Server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
HINFO	Host description	CPU and OS in ASCII
TXT	Text	Uninterpreted ASCII text

Domain_name **Time_to_live** **Class** **Type** **Value**

A portion of a possible DNS database for cs.vu.nl

; Authoritative data for cs.vu.nl

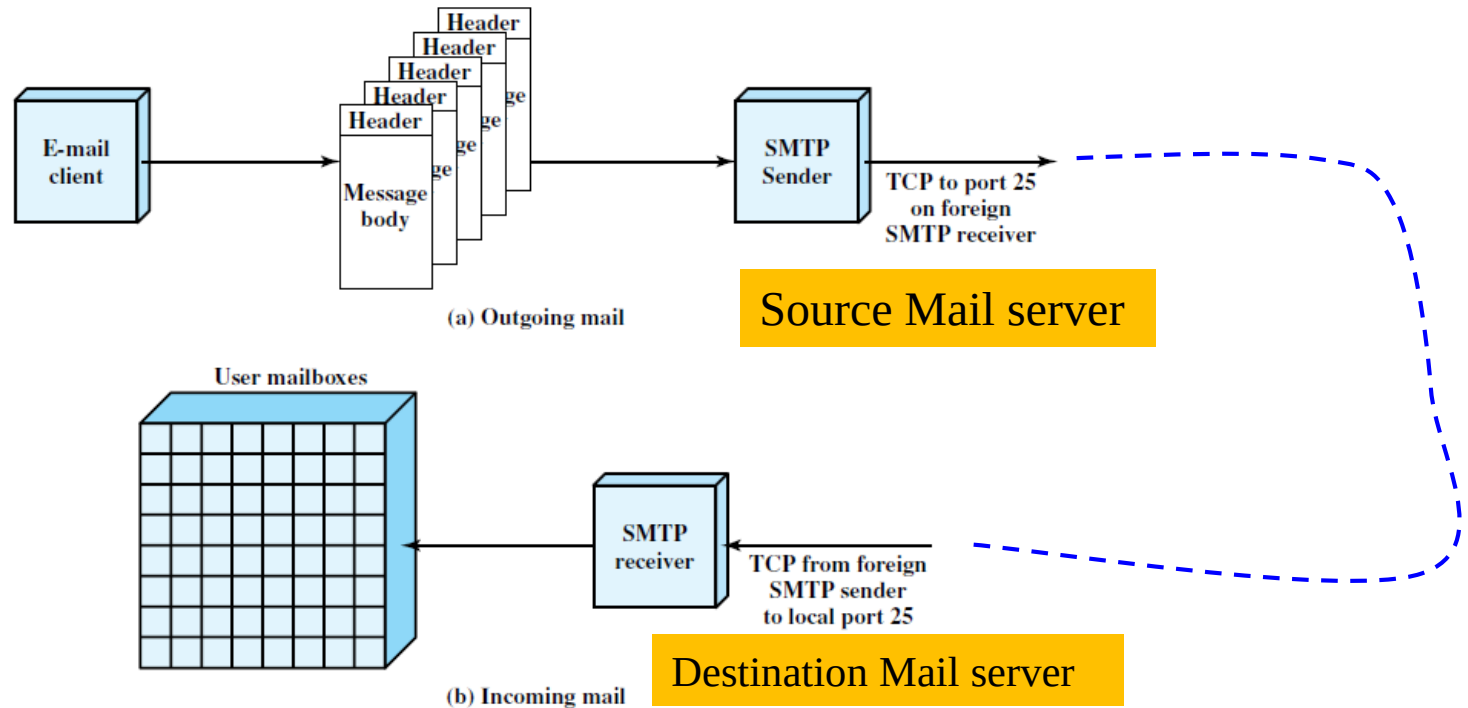
cs.vu.nl.	86400	IN	SOA	star boss (9527,7200,7200,241920,86400)
cs.vu.nl.	86400	IN	TXT	"Divisie Wiskunde en Informatica."
cs.vu.nl.	86400	IN	TXT	"Vrije Universiteit Amsterdam."
cs.vu.nl.	86400	IN	MX	1 zephyr.cs.vu.nl.
cs.vu.nl.	86400	IN	MX	2 top.cs.vu.nl.



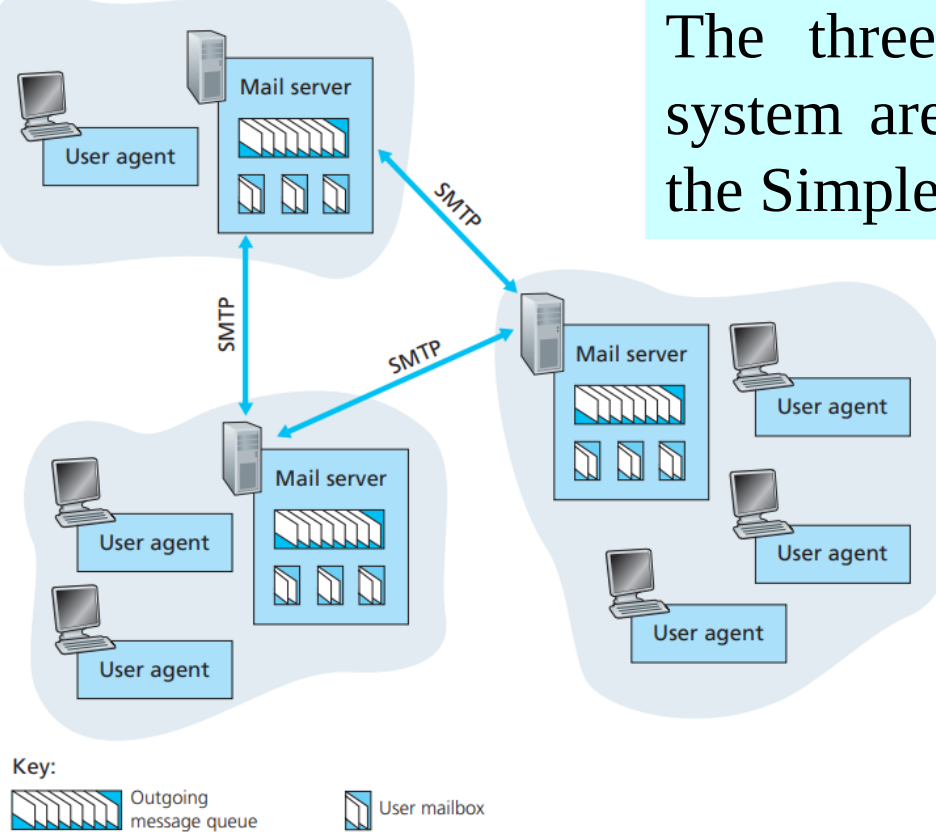
Two entries giving the first and second places to try to deliver e-mail sent to *person@cs.vu.nl*. The zephyr (a specific machine) should be tried first. If that fails, the top should be tried as the next choice.

SMTP-The Simple Mail Transfer Protocol

- ✓ The **Simple Mail Transfer Protocol (SMTP)** provides a basic electronic mail transport facility. Each created message consists of a header that includes the recipient's e-mail address and other information, and a body containing the message to be sent.
- ✓ This protocol (SMTP) transfers e-mail from the **mail server of a source** to the **mail servers of destinations** through TCP link.

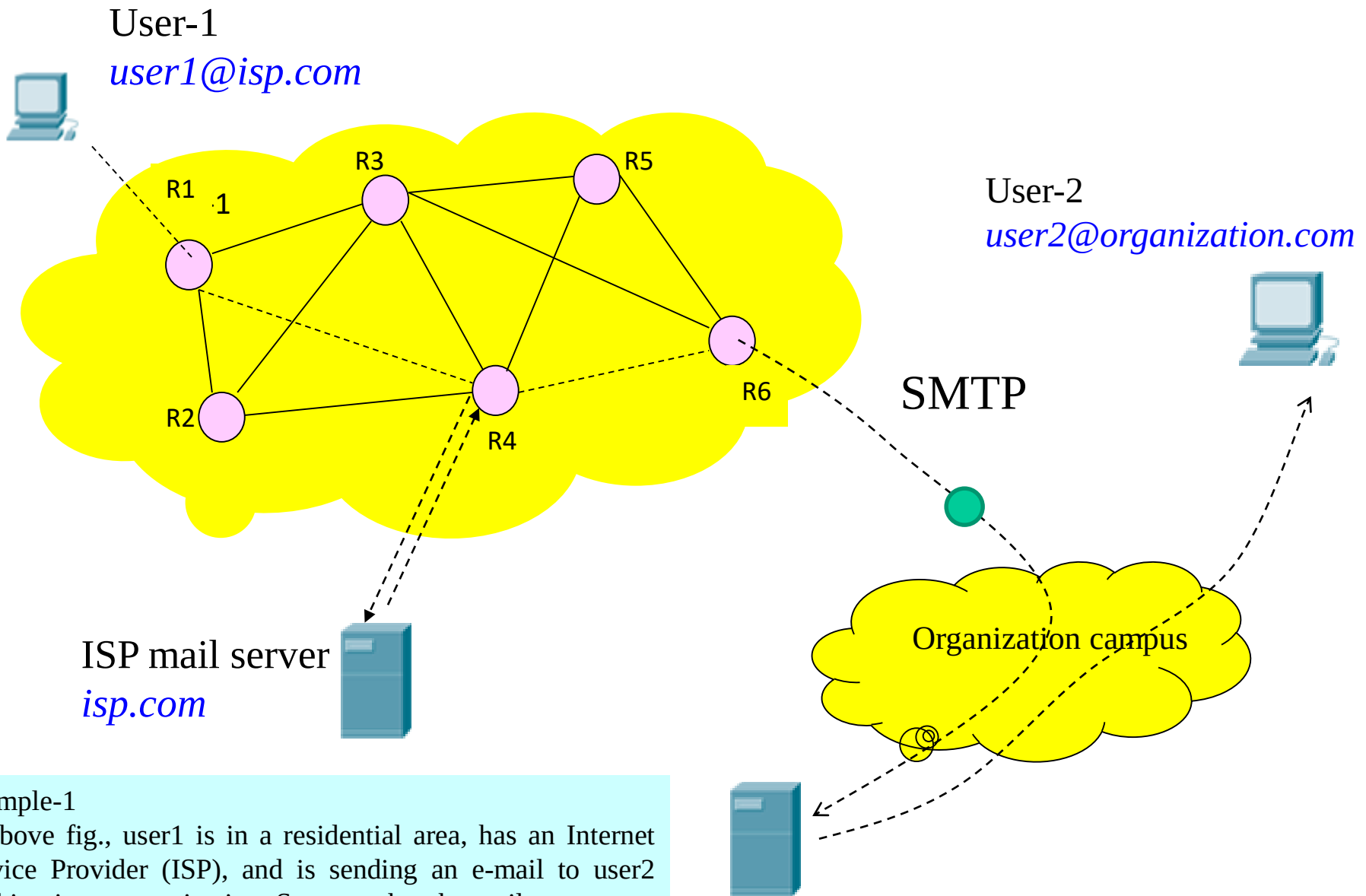


The three major components of e-mail system are: user agents, mail servers, and the Simple Mail Transfer Protocol (SMTP).



- ✓ Each recipient, such as Bob, has a mailbox located in one of the mail servers. Bob's mailbox manages and maintains the messages that have been sent to him.
- ✓ A **mailbox** is a space in the mail server allocated to the user to keep its e-mail

- ✓ In order for the mail to be sent properly from mailbox to mailbox, each mailbox must have a unique e-mail address.
- ✓ The e-mail address is multipart hierarchical address. The first portion of the address indicates the user and the later portion indicates the location of the mailbox.
- ✓ A typical message starts its journey in the sender's user agent, travels to the sender's mail server, and travels to the recipient's mail server, where it is deposited in the recipient's mailbox.



Example-1

In above fig., user1 is in a residential area, has an Internet Service Provider (ISP), and is sending an e-mail to user2 working in an organization. Suppose that the mail servers are *isp.com* and *organization.com* respectively. The e-mail addresses of user1 and user2 are *user1@isp.com* and *user2@organization.com* respectively.

Organization mail server
organization.com

The procedure for e-mail exchange between user 1 and user 2 are as follows.

1. User1 provides user2's e-mail address (*user2@organization.com*) and composes its message.
2. User1 sends the message to its mail server (*isp.com*)
3. Server *isp.com* places the message in its queue.
4. SMTP on user1's mail server notices the message in the queue and opens a TCP connection with the organization mail server (*organization.com*)
5. Initial handshaking takes place between the two servers.
6. The message is sent to the organization mail server using the established TCP connection.
7. User2's mail server receives the message and then puts it in user 2's mailbox, ready to be retrieved by user2.

A user mail box is a space in the mail server allocated to the user to keep its e-mail.

Post Office Protocol

- ✓ The most popular protocol used to transfer e-mail messages from a permanent **mailbox** to **local computer** is known as version 3 of the Post Office Protocol (POP3); a secure version of the protocol is known as POP3S.
- ✓ POP3 begins when the user agent (the client) opens a TCP connection to the mail server (the server) on port 110. The user first sends a login and password to authenticate the session.
- ✓ With the TCP connection established, POP3 progresses through three phases: authorization, transaction, and update.

- ✓ During the first phase, **authorization**, the user agent sends a username and a password (in the clear) to authenticate the user.
- ✓ During the second phase, **transaction**, the user agent retrieves messages; also during this phase, the user agent can mark messages for deletion, remove deletion marks, and obtain mail statistics.
- ✓ The third phase, **update**, occurs after the client has issued the quit command, ending the POP3 session; at this time, the mail server deletes the messages that were marked for deletion.

Mention the function of three phases of POP3.

Multipurpose Internet Mail Extensions (MIME)

The multipurpose Internet mail Extensions (MIME) were defined to allow transmission of non-ASCII data through e-mail. MIME does not change or replace protocols such as SMTP, POP3. Instead MIME allows arbitrary data to be encoded in ASCII and then transmitted in a standard e-mail message.

Fig. below illustrates a MIME message that contains a JPEG where the image is converted to 7-bit ASCII (base64 encoding) representation.

From: karim@juniv.edu

To: john@yahoo.com

MIME-Version:1.0

Content-Type: Image/jpeg

Content-Transfer-Encoding: base64

...data for the image...

Type of data in the message.

How the data in the message body is encoded?

- ✓ Each MIME message includes information that tells the recipient the type of data and the encoding used. A receiver's mail system must first convert from **base64** encoding back to binary, and then run an application that displays a JPEG image on the user's screen.

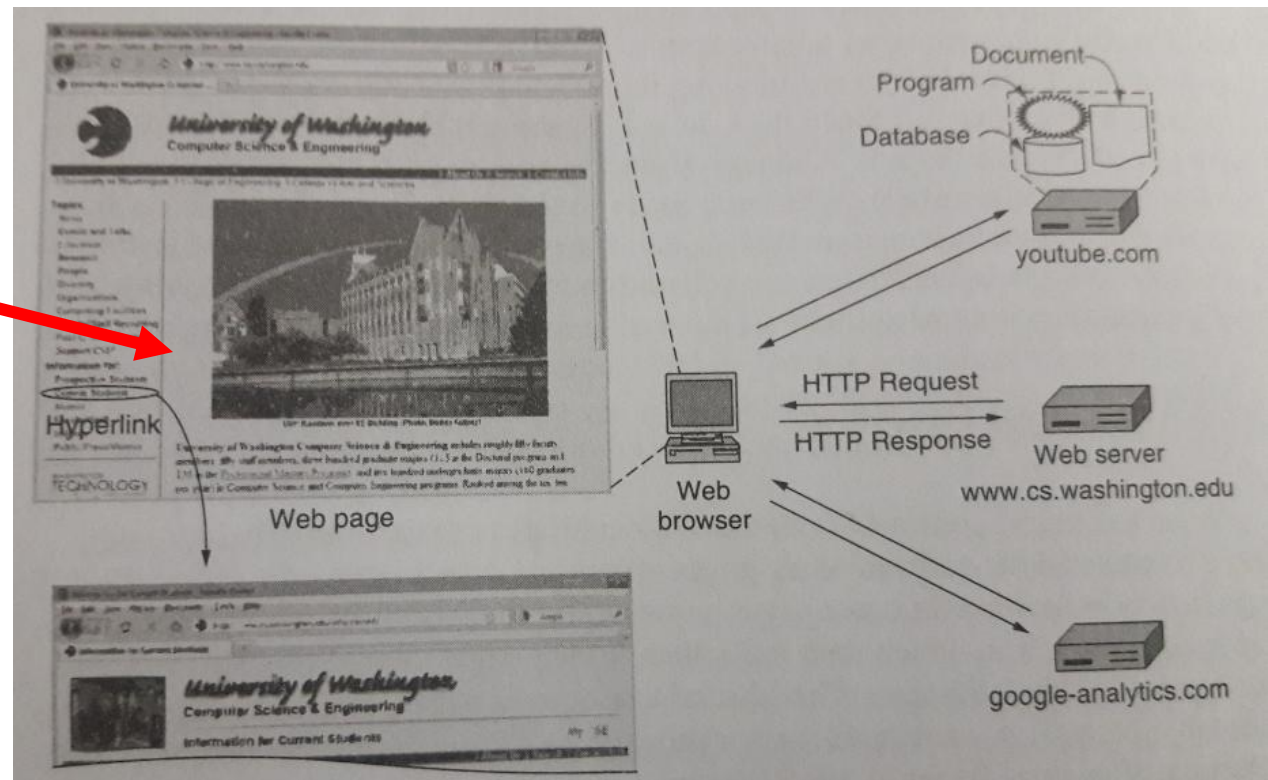
The World Wide Web

- ✓ Conceptually, *World Wide Web* (WWW) or simply *Web* consists of a vast, worldwide collection of contents in the form of **Web pages**, often called **pages** for short. The **Web pages** are accessible over the Internet.
- ✓ Each page may contain links to other pages anywhere in the world. The idea of having one page points to another called **hypertext**. A piece of text, icon, image and so on associated with another page is called a **hyperlink**.
- ✓ A Web page (also called a document) consists of objects. An object is simply a file—such as an HTML file, a JPEG image, a Java applet, or a video clip—that is addressable by a single URL.

✓ With a **web browser**, one can view **web pages** (**web page** may contain text, images, videos, and other multimedia) and navigate between them via hyperlinks.

- ✓The browser fetches the page requested, interprets the text and formatting commands (coding of web program) on it, and displays the page, properly formatted, on the screen.
- ✓A set of related web pages is called **website** like Jahangirnagar university website.

Web page may contain text, images, videos, and other multimedia.

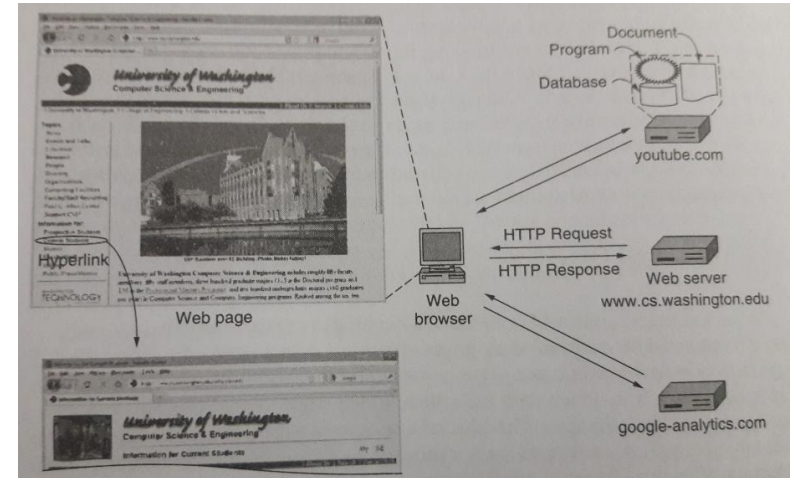


Client-Server Architecture

- ✓ In a **client-server architecture**, there is an **always-on** host, called the server, which services requests from many other hosts, called clients.
- ✓ Note that with the client-server architecture, clients do not directly communicate with each other; for example, in the Web application, two browsers do not directly communicate.

The Client Side

A **client host** requests an object (page) using browser and its URLs (Uniform Resource Locators). A typical URL is: *http://www.abcd.com/products.html*



✓ As an example, the URL of a page is:

http://www.cs.washington.edu/index.html, where the URL consists of three parts: the protocol (http), the DNS name of the host (www.cs.washington.edu), and the path name (index.html).

✓ Let us trace the steps that occur when our example link is selected.

Show three components of URL with example.

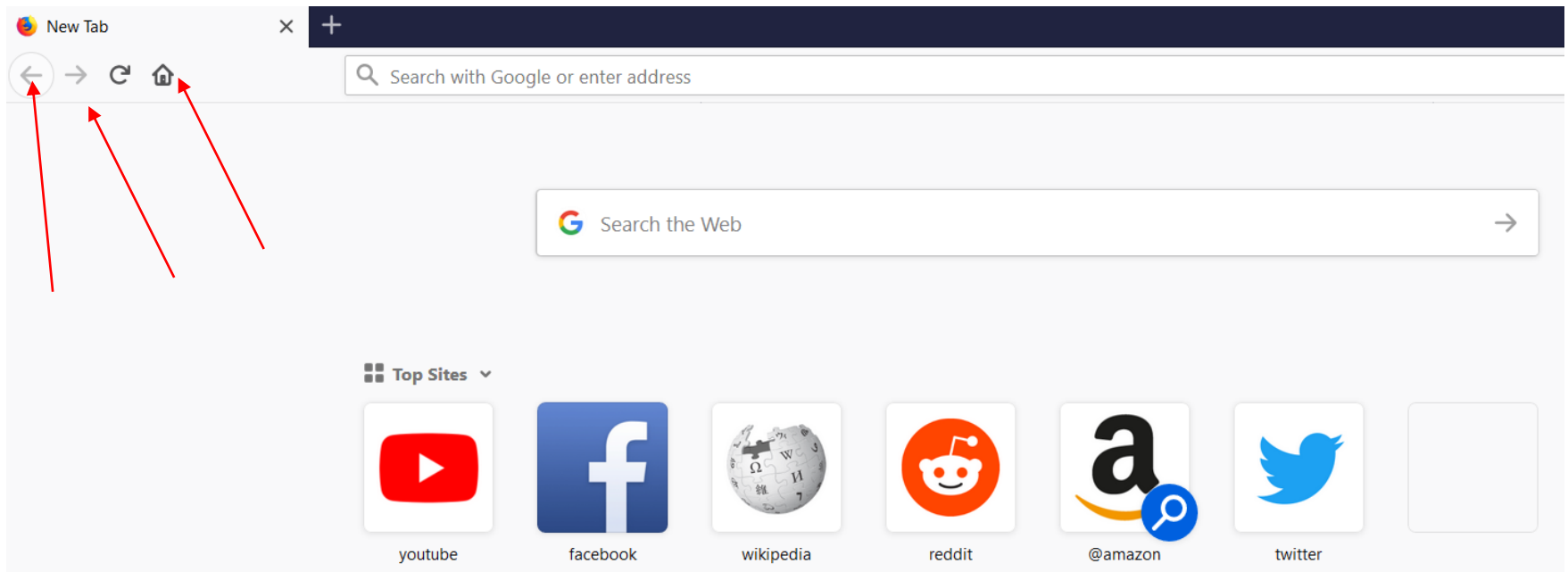
1. The browser determines the URL (by seeing what was selected).
2. The browser asks DNS server for the IP address of the server `www.cs.washington.edu`.
3. DNS server replies with `128.208.3.88`.
4. The browser makes a TCP connection to `128.208.3.88` on port 80, the well-known port for the HTTP protocol.
5. Client sends an HTTP request asking for the page `/index.html`.
6. The `www.cs.washington.edu` server sends the page as an HTTP response, for example, by sending the file `/index.html`.

7. If the page includes URLs that are needed for display, the browser fetches the other URLs using the same process. In this case, the URLs include multiple embedded images also fetched from `www.cs.washington.edu`, an embedded video from `youtube.com`, and a script from `google-analytics.com`.
7. The browser displays the page `/index.html`
8. The TCP connections are released if there are no other requests to the same servers for a short period.

Give the steps of client side of client server web browsing.

❑ Although a browser is basically an **HTML interpreter**, most browsers have numerous buttons and features to make it easier to navigate the Web.

❑ Most have a button for going back to the previous page, a button for going forward to the next page (only operative after the user has gone back from it), and a button for going straight to the user's own start page. For example the buttons of Mozilla Firefox is shown below.



The Server Side

- ✓ When the user types in a URL or clicks on a line of hypertext, the browser parses (breaks the components of URL) the URL and interprets the part between *http://* and the next slash as a DNS name to look up.
- ✓ Armed with the IP address of the server, the browser establishes a TCP connection to port 80 on that server. Then it sends over a command containing the rest of the URL, which is the name of a file on that server. The server then returns the file for the browser to display.

The steps that the server performs in its main loop are:

1. Accept a TCP connection from a client (a browser).
2. Get the name of the file requested.
3. Get the file (from disk).
4. Return the file to the client.
5. Release the TCP connection.

❖ A problem with this design is that every request requires making a disk access to get the file. The result is that the Web server cannot serve more requests per second than it can make disk accesses. A high-end SCSI (**Small Computer System Interface**) disk has an average access time of around 5 msec, which limits the server to at most 200 requests/sec.

❖ One obvious improvement (used by all Web servers) is to maintain a cache in memory of the n most recently used files. Before going to disk to get a file, the server checks the cache. If the file is there, it can be served directly from memory, thus eliminating the disk access.

URLs—Uniform Resource Locators

✓ If every page were somehow assigned a unique name, there would not be any ambiguity in identifying pages.

✓ URLs have three parts: the **protocol** (also known as the scheme), the **DNS name** of the machine on which the page is located, and a **local name** uniquely indicating the specific page (usually just a file name on the machine where it resides). As an example, the Web site for the author's department contains several videos about the university and the city of Amsterdam. The URL for the video page is

<http://www.cs.vu.nl/video/index-en.html>

✓ This URL consists of three parts: the protocol (*http*), the DNS name of the host (*www.cs.vu.nl*), and the file name (*video/index-en.html*), with certain punctuation separating the pieces. The file name is a path relative to the default Web directory at *cs.vu.nl*.

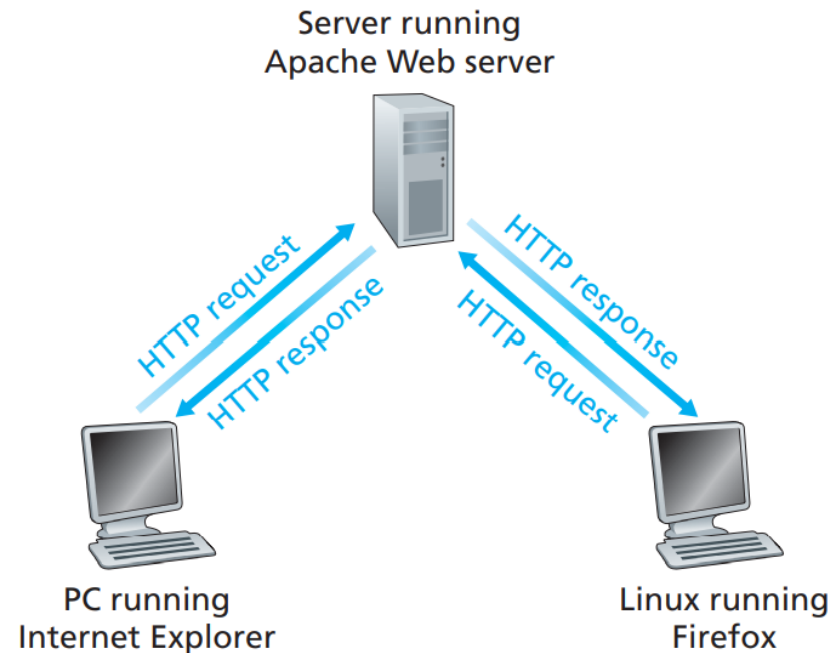
❖ URL scheme is open-ended in the sense that it is straightforward to have browsers use multiple protocols to get at different kinds of resources. In fact, URLs for various other common protocols have been defined. Slightly simplified forms of the more common ones are listed in Table-1

Table-1 Some common URLs

Name	Used for	Example
http	Hypertext (HTML)	http://www.ee.uwa.edu/~rob/
https	Hypertext with security	https://www.bank.com/accounts/
ftp	FTP	ftp://ftp.cs.vu.nl/pub/minix/README
file	Local file	file:///usr/suzanne/prog.c
mailto	Sending email	mailto:JohnUser@acm.org
rtsp	Streaming media	rtsp://youtube.com/montypython.mpg
sip	Multimedia calls	sip:eve@adversary.com
about	Browser information	about:plugins

HTTP Overview

- ✓ The protocol used for communication between a **browser** and a **web server** or between intermediate machines and web servers is known as HyperText Transfer protocol (HTTP).
- ✓ HTTP defines how Web clients request Web pages from Web servers and how servers transfer Web pages to clients. The general idea is illustrated in Figure below.



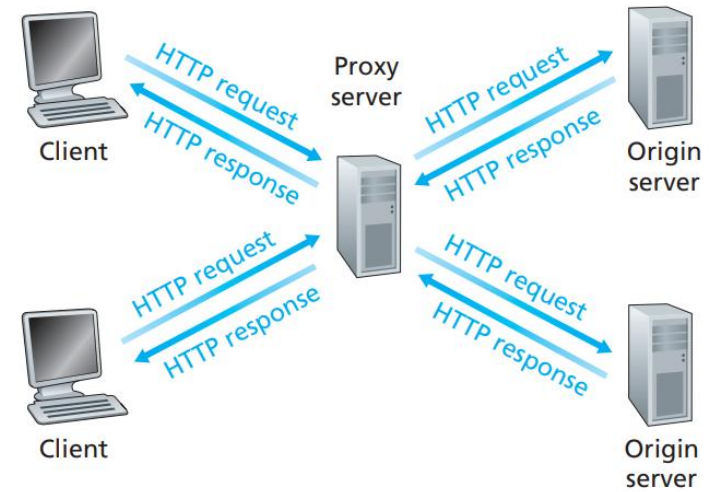
HTTP request-response behavior

- ✓ Because Web browsers (such as Internet Explorer and Firefox) implement the client side of HTTP, in the context of the Web, we will use the words browser and client interchangeably. Web servers, which implement the server side of HTTP, house Web objects, each addressable by a URL.
- ✓ The HTTP client first initiates a TCP connection with the server. Once the connection is established, the browser and the server processes access TCP through their socket interfaces.
- ✓ On the client side the socket interface is the door between the client process and the TCP connection; on the server side it is the door between the server process and the TCP connection.

- ✓ Once the TCP connection is in place, the browser and web server use HTTP to communicate; the browser sends a request to retrieve a specific page, and the server responds by sending a copy of the page.
- ✓ In most cases, a browser requests a Web page, and the server transfers a copy to the browser. HTTP also allows transfers from browser to server (form upload).

Web Caching

❖ A Web cache—also called a proxy server—is a network entity that satisfies HTTP requests on the behalf of an origin Web server. The Web cache has its own disk storage and keeps copies of recently requested objects in this storage.

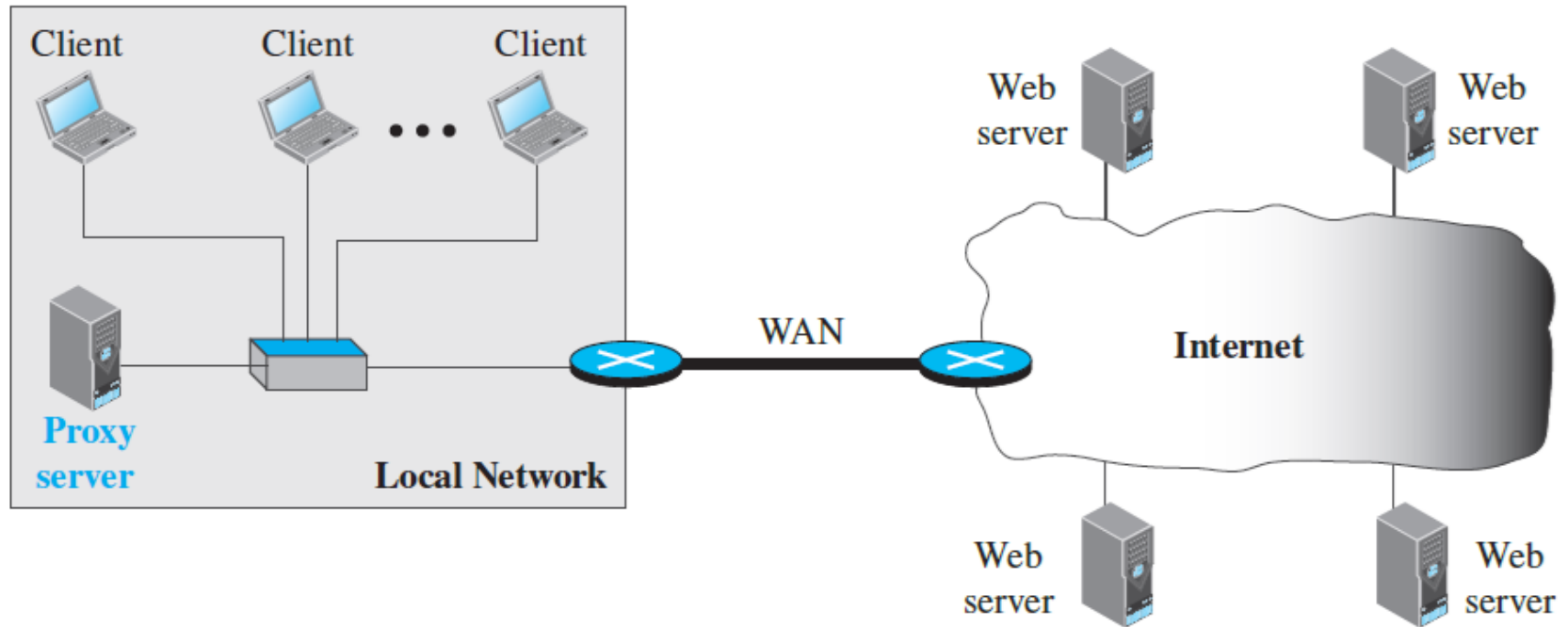


Proxy Server Location

The proxy servers are normally located at the client site. This means that we can have a hierarchy of proxy servers:

- ✓ A client computer can also be used as a proxy server, in a small capacity, that stores responses to requests often invoked by the client.
- ✓ In a company, a proxy server may be installed on the computer LAN to reduce the load going out of and coming into the LAN.
- ✓ An ISP with many customers can install a proxy server to reduce the load going out of and coming into the ISP network.

Figure below shows an example of a use of a proxy server in a local network, such as the network on a campus or in a company.



The caching process works as follows:

1. The browser establishes a TCP connection to the Web cache and sends an HTTP request for the object to the Web cache
2. The Web cache checks to see if it has a copy of the object stored locally called **cache hit**. If it does, the Web cache returns the object within an HTTP response message to the client browser.
3. If the resource is not cached, **cache miss** will result. Web cache opens a TCP connection to the origin server. The Web cache then sends an HTTP request for the object into the original server. After receiving this request, the origin server sends the object within an HTTP response to the Web cache.
4. When the Web cache receives the object, it stores a copy in its local storage and sends a copy, within an HTTP response message, to the client browser

Caching provides four benefits:

- a. Reduce load on the web server for example during admission test of Jahangir University.
- b. Reduce latency
- c. Reduce bandwidth utilization of users
- d. Network traffic reduction

Steps of web caching process and its benefits.

The three common type of cache is defined according to where the content is stored.

- **Browser cache** - this storage is done in the browser. All browsers have a local storage, which is usually used to retrieve previously accessed resources. This type of cache is private since stored resources are not shared.

- **Proxy cache** - this storage, also called intermediate caching, is done on the proxy server, between the client and the origin server. This is a type of shared cache as it's used by multiple clients and is usually maintained by ISP or the organization.

- **Reverse proxy cache** - closed to the web server maintained by the administrator.

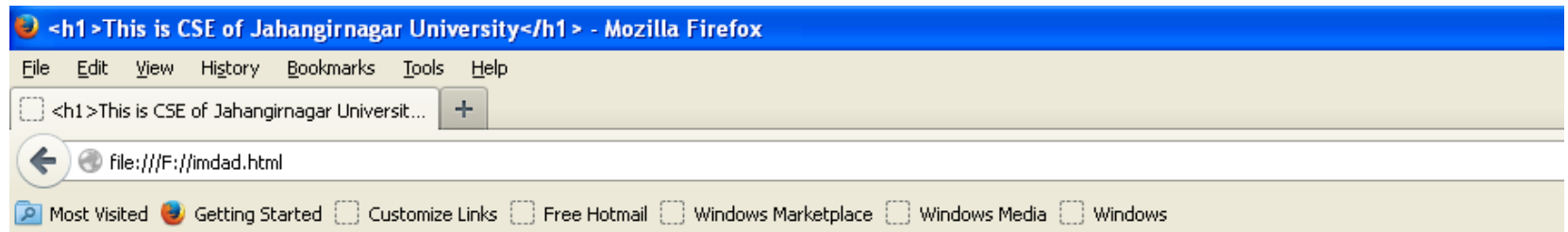
HTML-The HyperText Markup Language

- ✓ Web pages are written in a language called HTML (HyperText Markup Language). HTML allows users to produce Web pages that include text, graphics, and pointers to other Web pages.
- ✓ In the simplest form, web pages are **static**. That is, they are just files sitting on some server that present themselves in the same way each time they are fetched and viewed.
- ✓ HTML is a markup language, a language for describing how documents are to be formatted. The term "markup" comes from the old days when copyeditors actually marked up documents to tell the printer-in those days, a human being-which fonts to use, and so on.

✓Markup languages thus contain explicit commands for formatting. For example, in HTML, `<b>` means start boldface mode, and `</b>` means leave boldface mode. The advantage of a markup language over one with no explicit markup is that writing a browser for it is straightforward: the browser simply has to understand the markup commands.

Example

```
<html>
<head>
<title>
<h1>This is CSE of Jahangirnagar University</h1>
</title>
</head>
<body style="background-color:yellow;">
This is fourth year student <br/>
<i> Our favorite course is CSE-108 </i> <br/>
<h3> We will take thesis in next year </h3> <br/>
<p style="font-size:36px; text-align:center;">
Our M.Sc class will star soon </p>
<h3 style="font-family:arial;color:blue;"> We will take thesis in next year </h3> <br/>
</body>
</html>
```



This is fourth year student

Our favorite course is CSE-108

We will take thesis in next year

Our M.Sc class will star soon

We will take thesis in next year

Static Documents

- ✓ Static documents are fixed-content documents that are created and stored in a server. The client can get only a copy of the document. In other words, the contents of the file are determined when the file is created, not when it is used. Of course, the contents in the server can be changed, but the user cannot change them. Static website is created by using simple languages such as HTML, CSS, or JavaScript.

Dynamic Documents

- ✓ A dynamic document is created by a Web server whenever a browser requests the document. When a request arrives, the Web server runs an application program or a script that creates the dynamic document. The server returns the output of the program or script as a response to the browser that requested the document.
- ✓ A fresh document is created for each request, the contents of a dynamic document can vary from one request to another for example, stock prices, weather information, current availability of tickets for a concert, availability of product of e-commerce sites etc.
- ✓ In dynamic web pages, database is used.

CSS (Cascading Style Sheets) is used to represent appearance of pages.

PHP(Hypertext Preprocessor) for dynamic Web page for example e-commerce sites.

AJAX (Asynchronous JavaScript and XML) for full-featured and responsive Web applications: Google's Gmail, Maps etc. AJAX is a new technique for creating better, faster, and more interactive web applications with the help of XML, HTML, CSS, and Java Script/DOM.

Telnet and Secure Shell

- ✓ Telnet (teletype network) is a remote login protocol uses TCP/IP standard for establishing connection to a remote system. Recently IT specialists realize that telnet traffic is plain text hence suffers from security.
- ✓ To combat the situation, a new protocol SSH (or Secure Shell) is used to log into a remote machine which ensure the transmission of data thorough encrypted channel. We can use the SSH server to enable an SSH client for encrypted communications with Cisco port. Because of security the telnet protocol is replaced by Secure Shell (SSH) protocol.

File Transfer Protocol (FTP)

- ✓ File transfer is an important computer networking application. It is always essential that files and information geographically distributed over different location be shared among the numbers of working group.
- ✓ File Transfer Protocol (FTP) is part of the TCP/IP suite and is very similar to TELNET. Both FTP and TELNET are built on the client server paradigm, and both allow a user to establish a remote connection. However, TELNET provides a broader access to a user, whereas FTP allows access only to certain files. The sequence of operation of FTP is like:

1. A user requests a connection to a remote server
 2. The user waits for an acknowledgement.
 3. Once connected, the user must enter a user ID, followed by password.
 4. The connection is established over TCP session.
 5. The desired files are transferred.
 6. The user closes the FTP connection.
- FTP can also run through a Web browser.

Well-Known TCP Port Numbers

Ports are numbered and used as global standards to identify specific processes or types of network services provided at transport layer imposed from application layer. Ports indicate which service is desired. For example, TCP port 25 is for mail, and TCP port 80 is for HTTP.

20	FTP -- Data
21	FTP -- Control
22	SSH Remote Login Protocol
23	Telnet
25	SMTP
53	DNS
80	HTTP
110	POP3
161	SNMP

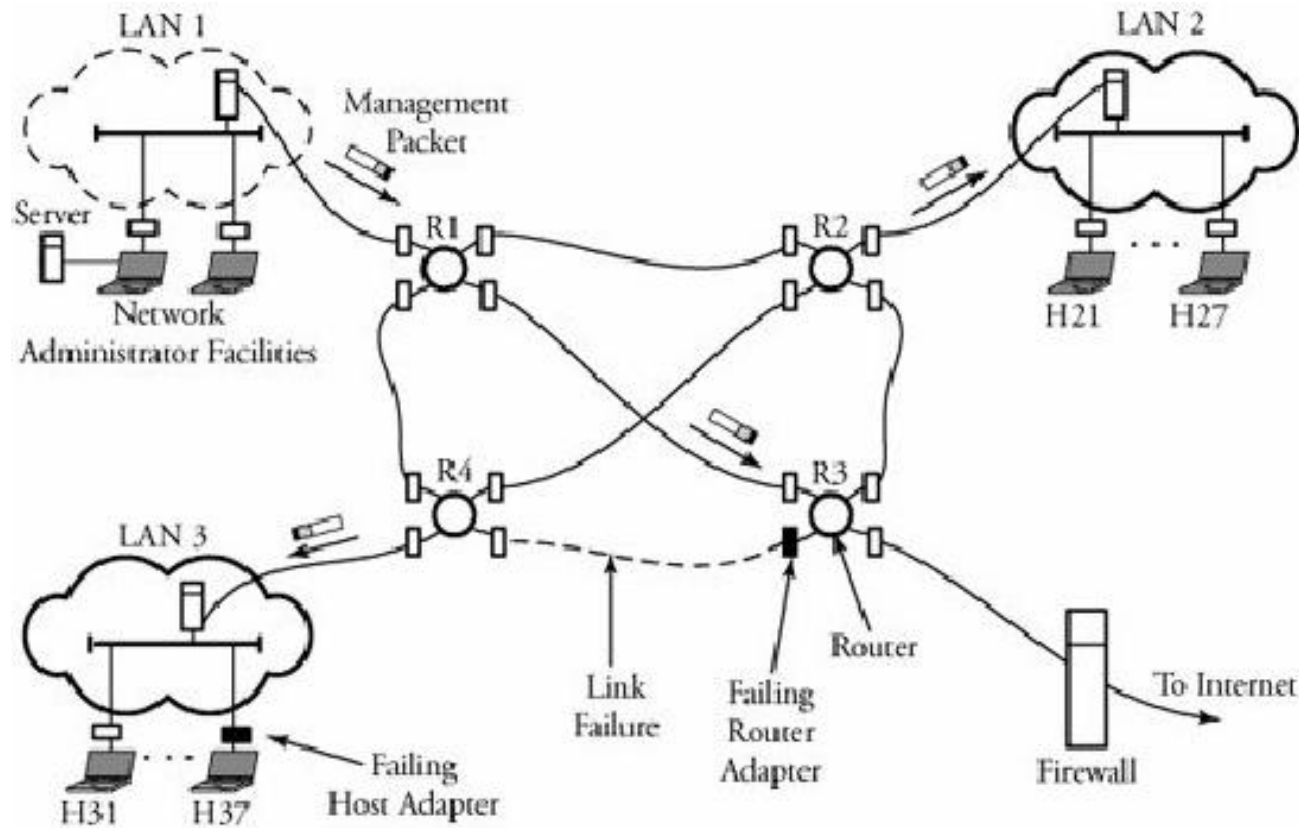
Network Management

- ✓ The main purpose of network management is to monitor, manage, and control a network. A network can be structured with many **links, routers, switches, servers, and other physical-layer devices**, which can be equipped with many network protocols that coordinate them.
- ✓ Imagine when thousands of such devices or protocols are tied together by an ISP/organization network and how drastic their management can become to avoid any interruptions in routine services.
- ✓ In this context the purpose of **network management** is to monitor, test, and analyze the hardware and software of components of the entire network.

Network management tasks can be characterized as follows:

QoS and performance management. A network administrator periodically monitors and analyzes routers, hosts, and utilization of links and then redirect traffic flow to avoid any overloaded spots. Certain tools are available to detect rapid changes in traffic flow.

Network failure management. Any fault in a network, such as link, host, or router hardware or software outages, must be detected, located, and responded to by the network. Typically, increased checksum errors in frames is an indication of possible error. Figure shows adapter (ethernet port fa0/0) failures at router R3 and host H37; these failures can be detected through network management.



Simple network management in a scenario of LANs connecting to the Internet

Configuration management. This task involves tracking all the devices under management and ensuring that all devices are connected and operate properly. If there is an unexpected change in routing tables, a network administrator wants to discover the misconfigured spot and reconfigure the network before the error affects the network substantially.

Security management. A network administrator is responsible for the security of its network. This task is handled mainly through firewalls. A firewall can monitor and control access points.

Billing and accounting management. The network administrator specifies user access or restrictions to network resources and issues all billing and charges, if any, to users.

BW management, for example using MikroTik router.